

M1: See how typos cause search failures

Beyond Benchmarks: Robustness for Real-World Information Systems

Bum Jun Kim, The University of Tokyo, CIKM 2026

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Learning objective: Find out whether search missed a relevant paper or reranking placed it too low. Check which papers reach the LLM.

Run this notebook on its own with Restart Kernel → Run All. See the README for setup.

```
# Load the libraries, SciFact data, and saved model results.
from pathlib import Path
import sys
ROOT = next(p / "resources" for p in [Path.cwd(), *Path.cwd().parents] if (p / "resources" /
"tutorial_utils").is_dir())
if str(ROOT) not in sys.path: sys.path.insert(0, str(ROOT))
from IPython.display import display, Markdown

from tutorial_utils.presentation import install_plain_display
install_plain_display()

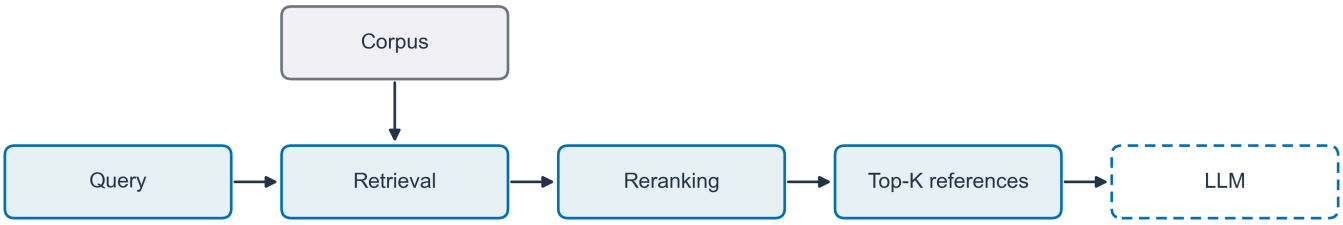
import pandas as pd
pd.set_option("display.max_colwidth", 120)
from tutorial_utils import load_tutorial_context, retrieve, rerank
from tutorial_utils import observation as obs
from tutorial_utils import plotting as viz
MODE = "cached"  # Use "live" for new model inference; see README for setup.
ctx = load_tutorial_context(mode=MODE, seed=42)
print(ctx.banner())
```

```
MODE=cached | SciFact | 5183 documents | 300 test queries
BM25 / MiniLM dense exact retrieval → MiniLM cross-encoder (20 candidates)
```

From a query to an LLM answer

Search finds papers for the LLM to use as references. This notebook runs search and builds the prompt. The dashed LLM step shows where answer generation would happen.

```
# Show the pipeline from a search query to an LLM answer.
viz.show(viz.pipeline_figure(), description="The corpus supplies candidate papers; selected references support
answer generation.")
```



Dense search over all papers → 20 candidates → cross-encoder reranking → K references (K=5 here) → LLM answer.

Interpretation: The LLM sees only the K references, so a relevant paper that search misses or reranking places below rank K never reaches the answer.

A candidate is a paper selected by search. Reranking scores the 20 candidates again and sorts them. BM25 is a separate search method that matches words.

Typos change the query. Document attacks add a passage to the collection. M2 examines both changes. M3 measures the loss, M4 tests typo repairs, M5 filters attack documents, and M6 checks whether a repair is ready to use.

Setup: SciFact and saved results

Keep `resources/` next to the notebooks. The default cached mode runs offline on CPU. Dense search uses saved text vectors. BM25 and reranking use saved rankings.

Optional `MODE="live"` computes new query vectors and candidate scores, using the saved document vectors. See the README for setup and cache help. Cached timings show local processing time; they do not show the time needed to run the models.

SciFact contains 5,183 papers and 300 official test queries. Each query is a scientific claim. Each paper has a title and abstract. Training and validation queries come from the separate official train set.

A qrel is a recorded relevance label for a query–paper pair. A relevant paper can support or refute the claim. We check whether search finds these papers. We do not decide whether the claim is true.

What robustness means here

A robust search system finds relevant evidence even when the query has a typo. It also works well on correctly written queries.

We compare each original query with its typo version. The papers, models, text processing, and qrels stay fixed. We measure both relevance and ranking changes. Identical result lists can still miss all the evidence.

Experiment Settings: Query and maximum edits

Choose a query and the maximum number of character edits. Then select the code cell below and use Run All Below. M3 tests all edit settings on all 300 queries.

```
# Choose a query and the maximum number of character edits.
query_id = "1303"  # Tirasemtiv (default); "1012": radioiodine; "628": HTLV-1.
severity = 1       # Maximum character edits: 0 (no edits), 1, 2, or 3.
```

Can one typo remove all relevant papers from the top 10?

The LLM reads only the papers selected for its prompt. If search misses the evidence, the LLM cannot use it. We inspect this failure with SciFact.

Dense search: Compare text vectors

The `all-MiniLM-L6-v2` encoder turns each query and paper into a vector of 384 numbers. It processes the query and paper separately. Its trained weights stay fixed.

The encoder reads up to 256 tokens; a token can be part of a word. It averages the final token vectors, ignores padding, and scales the result to length 1. This scaling is called L2 normalization. See the encoder's pooling example.

$$\cos(u, v) = \frac{u^\top v}{\|u\|_2 \|v\|_2}, \qquad s(q, d) = \cos(z_q, z_d) = z_q^\top z_d.$$

Cosine similarity compares vector directions. Its value ranges from −1 to 1. Because these vectors have length 1, their dot product gives the same score. Search ranks higher scores first. A high score does not guarantee relevance. We can compute paper vectors once and reuse them.

The cross encoder reads the query and each candidate together. It gives each pair a relevance score, called a logit, and sorts the candidates by that score. The logit is the output of the reranker's learned scoring head, not cosine similarity. The reranker was trained on MS MARCO passage ranking. The sentence encoder learned from a broader set of sentence pairs.

The prompt contains the claim, answer instructions, and selected papers. Using retrieved documents to help an LLM answer is called retrieval-augmented generation (RAG).

Three search methods

- BM25 matches words. Rare words get more weight. The score also accounts for repeated words and document length.
- Dense search compares query and paper vectors.
- Dense → reranker retrieves candidates with dense search, then scores each query–paper pair together.

Our BM25 ignores case and counts each distinct query word once. It uses $k_1 = 1.2$ and $b = 0.75$. It keeps common words and does not shorten words to their stems.

The sentence encoder learned from sentence pairs, including scientific-paper pairs. The cross encoder reranks the candidates. Both models keep their trained weights.

The next table shows the first five reranked papers for the original query.

```
# Show dataset sizes and the original query's top five reranked papers.
print(f"Documents: {len(ctx.corpus):,} | official BEIR test queries: {len(ctx.test_ids)}")
print(f"Derived train: {len(ctx.train_ids)} | validation: {len(ctx.val_ids)}")
# Compare BM25 with dense search and reranking on the original query.
# Cached mode uses saved model results.
baseline = obs.observe(ctx, query_id, severity=0)
obs.show_baseline(ctx, baseline)
```

```
Documents: 5,183 | official BEIR test queries: 300
Derived train: 647 | validation: 162
```

Original query: Tirasemtiv has no effect on fast-twitch muscle.

Original query: first five reranked papers (one query, full collection)

Rank	Document ID	Logit	Judgment	Title
1	12631697	-4.679	judged relevant	Activation of fast skeletal muscle troponin as a potential therapeutic approach for treating neuromuscular diseases
2	40631095	-6.432	unjudged	The Combination of Exercise and Respiratory Training Improves Respiratory Muscle Function in Pulmonary Hypertension
3	23737024	-6.775	unjudged	Acute exercise activates nuclear factor (NF)-kappaB signaling pathway in rat skeletal muscle.
4	34328964	-6.877	unjudged	Intimal smooth muscle cell proliferation after balloon catheter injury. The role of basic fibroblast growth factor.
5	37037012	-7.002	unjudged	Regional vascular resistance during exercise: role of cardiac afferents and exercise training.

These five references include 1 of 1 known relevant paper, at rank 1.

Rank 1 is first after reranking. Document ID identifies the paper. Logit is the model score for the query–paper pair. The cross encoder reads the claim and the paper's title and abstract together as one input of up to 512 tokens, and its final layer outputs one raw number. Higher ranks earlier. It is not a probability: the number is never converted to a 0–1 scale, so it can be negative, and only its order among the 20 candidates matters. Judgment is the qrel label. Title is the full paper title.

Unjudged means this query–paper pair has no relevance label. It gets zero credit in evaluation; its relevance is unknown.

Interpretation: With the original query, the reranker places the only known relevant paper first, so the LLM would read it as Reference 1.

What goes into the LLM prompt?

`rag_messages` contains two messages:

- The system message gives the answer instructions.
- The user message gives the original claim and five numbered references. Each reference includes its SciFact ID, full title, and full abstract.

Qrels are used only for evaluation. They are not included in the prompt.

K counts papers. The LLM's context limit counts tokens. The prompt and the answer must fit within that limit. Here the prompt uses each paper's full title and abstract, even though the search encoder reads at most 256 tokens. M5 shows how attacks and filtering change the references.

```python
# Build and display the LLM input using the top K reranked papers.
K = 5
rag_candidates = retrieve(ctx, [query_id], method="dense", top_k=20, severity=0)
rag_ranking = rerank(ctx, rag_candidates, top_k=20, severity=0)[query_id]
rag_reference_ids = [doc_id for doc_id, _ in rag_ranking[:K]]
reference_blocks = []
for number, doc_id in enumerate(rag_reference_ids, start=1):
    title, _, abstract = ctx.corpus[doc_id].partition("\n")
    reference_blocks.append(
        f"[Reference {number} | Document ID: {doc_id}]\n\n"
        f"Title: {title}\n\nAbstract: {abstract}"
    )

rag_messages = [
    {
        "role": "system",
        "content": (
            "Assess whether the scientific claim is supported, refuted, or not established "
            "by the reference documents. Use only these references, cite their reference "
            "numbers, and state when the evidence is insufficient."
        ),
    },
    {
        "role": "user",
        "content": (
            f"Claim: {ctx.queries[query_id]}\n\nReference documents:\n\n"
            + "\n\n".join(reference_blocks)
        ),
    },
]
print(f"LLM input preview: selected test query, clean input, K={K} references.")
for message in rag_messages:
    display(Markdown(f"#### {message['role'].capitalize()} message\n\n" + message["content"]))
obs.show_prompt_interpretation(ctx, query_id, rag_reference_ids)
```

```
LLM input preview: selected test query, clean input, K=5 references.
```

System message

Assess whether the scientific claim is supported, refuted, or not established by the reference documents. Use only these references, cite their reference numbers, and state when the evidence is insufficient.

User message

Claim: Tirasemtiv has no effect on fast-twitch muscle.

Reference documents:

[Reference 1 | Document ID: 12631697]

Title: Activation of fast skeletal muscle troponin as a potential therapeutic approach for treating neuromuscular diseases

Abstract: Limited neural input results in muscle weakness in neuromuscular disease because of a reduction in the density of muscle innervation, the rate of neuromuscular junction activation or the efficiency of synaptic transmission. We developed a small-molecule fast-skeletal-troponin activator, CK-2017357, as a means to increase muscle strength by amplifying the response of muscle when neural input is otherwise diminished secondary to neuromuscular disease. Binding selectively to the fast-skeletal-troponin complex, CK-2017357 slows the rate of calcium release from troponin C and sensitizes muscle to calcium. As a consequence, the force-calcium relationship of muscle fibers shifts leftwards, as does the force-frequency relationship of a nerve-muscle pair, so that CK-2017357 increases the production of muscle force in situ at sub-maximal nerve stimulation rates. Notably, we show that sensitization of the fast-skeletal-troponin complex to calcium improves muscle force and grip strength immediately after administration of single doses of CK-2017357 in a model of the neuromuscular disease myasthenia gravis. Troponin activation may provide a new therapeutic approach to improve physical activity in diseases where neuromuscular function is compromised.

[Reference 2 | Document ID: 40631095]

Title: The Combination of Exercise and Respiratory Training Improves Respiratory Muscle Function in Pulmonary Hypertension

Abstract: Increased dyspnea and reduced exercise capacity in pulmonary arterial hypertension (PAH) can be partly attributed to impaired respiratory muscle function. This prospective study was designed to assess the impact of exercise and respiratory training on respiratory muscle strength and 6-min walking distance (6MWD) in PAH patients. Patients with invasively confirmed PAH underwent 3 weeks of in-hospital exercise and respiratory training, which was continued at home for another 12 weeks. Medication remained constant during the study period. Blinded observers assessed efficacy parameters at baseline (I) and after 3 (II) and 15 weeks (III). Respiratory muscle function was assessed by twitch mouth pressure (TwPmo) during nonvolitional supramaximal magnetic phrenic nerve stimulation. Seven PAH patients (4 women; mean pulmonary artery pressure 45 ± 11 mmHg, median WHO functional class 3.1 ± 0.4, idiopathic/associated PAH n = 5/2) were included. The training program was feasible and well tolerated by all patients with excellent compliance. TwPmo was I: 0.86 ± 0.37 kPa, II: 1.04 ± 0.29 kPa, and III: 1.27 ± 0.44 kPa, respectively. 6MWD was I: 417 ± 51 m, II: 509 ± 39 m, and III: 498 ± 39 m, respectively. Both TwPmo (+0.41 ± 0.34 kPa, +56 ± 39 %) and 6MWD (+81 ± 30 m, +20 ± 9 %) increased significantly in the period between baseline and the final assessment (pairwise comparison: p = 0.012/<0.001; RM-ANOVA considering I, II, III: p = 0.037/<0.001). Exercise and respiratory training as an adjunct to medical therapy may be effective in patients with PAH to improve respiratory muscle strength and exercise capacity. Future, randomized, controlled trials should be carried out to further investigate these findings.

[Reference 3 | Document ID: 23737024]

Title: Acute exercise activates nuclear factor (NF)-kappaB signaling pathway in rat skeletal muscle.

Abstract: Two studies were performed to investigate the effects of an acute bout of physical exercise on the nuclear protein kappaB (NF-kappaB) signaling pathway in rat skeletal muscle. In Study 1, a group of rats (n=6) was run on the treadmill at 25 m/min, 5% grade, for 1 h or until exhaustion (Ex), and compared with a second group (n=6) injected with two doses of pyrrolidine dithiocarbamate (PDTC, 100 mg/kg, i.p.) 24 and 1 h prior to the acute exercise bout. Three additional groups of rats (n=6) were injected with either 8 mg/kg (i.p.) of

lipopolysaccharide (LPS), 1 mmol/kg (i.p.) t-butylhydroperoxide (tBHP), or saline (C) and killed at resting condition. Ex rats showed higher levels of NF-kappaB binding and P50 protein content in muscle nuclear extracts compared with C rats. Cytosolic IkappaBalpha and IkappaB kinase (IKK) contents were decreased, whereas phospho-IkappaBalpha and phospho-IKK contents were increased, comparing Ex vs. C. The exercise-induced activation of NF-kappaB signaling cascade was partially abolished by PDTC treatment. LPS, but not tBHP, treatment mimicked and exaggerated the effects observed in Ex rats. In Study 2, the time course of exercise-induced NF-kappaB activation was examined. Highest levels of NF-kappaB binding were observed at 2 h postexercise. Decreased cytosolic IkappaBalpha and increased phosphor-IkappaBalpha content were found 0-1 h postexercise whereas P65 reached peak levels at 2-4 h. These data suggest that the NF-kappaB signaling pathway can be activated in a redox-sensitive manner during muscular contraction, presumably due to increased oxidant production. The cascade of intracellular events may be the overture to elevated gene expression of manganese superoxide dismutase reported earlier (Pfluegers Arch. 442, 426-434, 2001).

[Reference 4 | Document ID: 34328964]

Title: Intimal smooth muscle cell proliferation after balloon catheter injury. The role of basic fibroblast growth factor.

Abstract: The localization and synthesis of basic fibroblast growth factor (bFGF) in the rat carotid artery were investigated at times of chronic smooth muscle cell proliferation. Immunocytochemical staining showed the presence of bFGF in the uninjured arterial wall, and after balloon injury, this cellular staining was decreased. Western and northern blot analyses likewise showed that the amount of bFGF protein and mRNA decreased after injury. A neutralizing antibody to bFGF was administered 4 and 5 days after injury and was found to have no effect on intimal smooth muscle cell proliferation. These data suggest that an increase in the expression of bFGF is not necessary for chronic smooth muscle cell proliferation observed after balloon catheter injury and that bFGF is not the major mitogen responsible for intimal smooth muscle cell proliferation.

[Reference 5 | Document ID: 37037012]

Title: Regional vascular resistance during exercise: role of cardiac afferents and exercise training.

Abstract: This study was designed to determine whether cardiac vagal afferents exert an inhibitory influence on increases in regional vascular resistance during exercise and to determine whether endurance exercise training enhances the inhibitory influence of cardiac vagal afferents. We measured changes in regional vascular resistance in 12 rabbits at rest and during running at 12.6 m/min, 20% grade, before and after reversible denervation of cardiac afferents (intrapericardial procainamide HCl, 2%). In addition, these procedures were repeated in five of these rabbits following an 8-wk endurance exercise training program. Because intrapericardial injections of procainamide anesthetize both the efferent as well as the afferent innervation to the heart, it was necessary to determine the effects of blocking the efferent innervation on the regulation of regional vascular resistance during exercise. Rabbits were instrumented with Doppler ultrasonic flow probes around the renal (R), mesenteric (M), ascending, and terminal aortic (TA) arteries. Catheters were positioned in the central ear artery and vein and pericardial sac. Mean arterial pressure, heart rate, cardiac output, R, M, TA, and systemic (S) resistances were determined. Exercise changed R (+37 +/- 4%), M (+88 +/- 9%), TA (-62 +/- 6%), and S (-34 +/- 3) resistances. Subsequent cardiac efferent blockade alone had no significant effect on regional vascular resistance during exercise. Combined efferent and afferent blockade resulted in significant increases in R (+62 +/- 6%) and M resistance (+134 +/- 13%) but did not alter TA (-51 +/- 4%) or S (-27 +/- 2%) resistance during exercise. Exercise training significantly enhanced the inhibitory influence of cardiac afferents on R and M regional vascular resistance.(ABSTRACT TRUNCATED AT 250 WORDS)

Interpretation: The LLM sees only the claim, the instructions, and these 5 papers; the known relevant paper is Reference 1, so the evidence is in its input.

Search quality: Recall@10 and nDCG@10

Recall@10 is the fraction of known relevant papers found in the first ten results. nDCG@10 also rewards placing them earlier. Both range from 0 to 1; higher is better.

If there is one relevant paper, rank 1 and rank 2 both give Recall@10=1. nDCG@10 is 1 at rank 1 and about 0.631 at rank 2. A paper below rank 10 adds nothing to either score.

nDCG divides the ranking's score by the best possible score for that query. The appendix shows how. All 300 test queries have at least one known relevant paper.

The next table uses the original query. Later, ΔnDCG@10 = typo score − original score. Negative means a loss; positive means a gain. Zero means equal scores, even if the returned papers differ.

```
# Compare all three systems on the original query.
obs.show_comparison(obs.observe(ctx, query_id, severity=0), clean_only=True)
```

Query: Tirasemtiv has no effect on fast-twitch muscle. — one query, full collection, top 10 scored, original input

System	Original nDCG@10	Original Recall@10
BM25	1.000	1.000
Dense	1.000	1.000
Dense → reranker	1.000	1.000

All three systems score 1 on nDCG@10 and Recall@10 for this original query.

Dense candidate Recall@20: original 1.000; reranked Recall@10: original 1.000 (both higher is better).

Interpretation: Every system already ranks the known evidence as well as possible, so any later loss on this claim comes from the typo.

Reranking can move a candidate, but cannot add a missing paper

Each example has one known relevant paper. Dense search gives 20 candidates to the reranker. The LLM receives the first five reranked papers.

Relevant paper: dense rank → reranked rank	Candidate Recall@20	Recall@10: before → after	nDCG@10: before → after	Included at K=5: before → after
15 → 3	1	0 → 1	0 → 0.500	No → Yes
8 → 2	1	1 → 1	0.315 → 0.631	No → Yes
Not retrieved → still missing	0	0 → 0	0 → 0	No → No

Interpretation: Reranking can move a candidate into the top ten or the prompt (first two rows), but it cannot recover a paper missing from the candidates (last row).

Candidate Recall@20 is the fraction of known relevant papers among the 20 candidates. If two papers are relevant and search finds one, it is 0.5. Reranking can move that paper up, but final Recall@10 cannot exceed 0.5. nDCG also rewards earlier ranks. Higher is better for every recall and nDCG column.

Did search miss the evidence, or did reranking place it too low?

Compare the same claim before and after a typo. First check whether the relevant paper is among the 20 candidates. If it is missing, search failed. If it is present but ends up too low, check reranking.

To measure how much the typo changed the result, subtract the claim's original nDCG@10 from its typo nDCG@10. This is the ΔnDCG@10 shown in the tables:

$$\Delta_q = \mathrm{nDCG@10}(\tilde{q}) - \mathrm{nDCG@10}(q).$$

Here q is the original claim and $\tilde{q}$ is the same claim after the typo. The papers, models, and qrels stay fixed, so Δ_q reflects only the typo. $\Delta_q < 0$ means the typo lowered the score, $\Delta_q = 0$ means no change, and $\Delta_q > 0$ means a gain. Because nDCG@10 lies between 0 and 1, Δ_q lies between −1 and 1; $\Delta_q = -1$ means a perfect top ten lost every known relevant paper.

The three queries come from M3's Selected severity-1 cases table. With one edit, their Dense nDCG@10 scores change as follows:

- Tirasemtiv (`1303`): 1.0000 → 0.0000, so $\Delta_q = -1$.
- Radioiodine (`1012`): 1.0000 → 1.0000, so $\Delta_q = 0$.
- HTLV-1 (`628`): 0.4307 → 1.0000, so $\Delta_q = +0.5693$.

Interpretation: A single character edit can erase a claim's score, leave it unchanged, or even raise it, so one claim's Δ_q cannot show how typos affect search in general.

These scores use `dense_s0` , `dense_s1` , and the official test qrels. The rankings are saved in `resources/assets/caches/runs.json` . We chose the cases after scoring the test set to show a loss, no change, and a gain. Use M3's full-test counts to see how often each occurs.

Similar rankings can still miss the evidence

Each typo edit inserts, deletes, or replaces one character in an eligible word. The edits are fixed, so rerunning gives the same result. The rules protect digits, listed negation words, and many identifier-like tokens.

`severity` is the maximum number of edits. Some queries have too few eligible words to reach it.

Finite RBO@10 compares the paper IDs and order in two top-ten lists. It does not use qrels. The score is 0 for no overlap and 1 for identical ordered lists. Higher means more agreement, but both lists can still miss every relevant paper. Different lists can also have equal nDCG.

Optional calculation: $S_{1:d}$ and $T_{1:d}$ contain the first d papers from each list. The fraction they share is $A_d = |S_{1:d} \cap T_{1:d}|/d$. With $p = 0.9$:

$$\mathrm{RBO}_{10,0.9} = \frac{\sum_{d=1}^{10}(1-0.9)0.9^{d-1}A_d}{1-0.9^{10}}.$$

Earlier ranks receive more weight. The denominator scales the total weight to 1. This follows Webber, Moffat & Zobel's rank-overlap framework (2010).

```
# Show the original and typo queries with their search results.
demo = obs.observe(ctx, query_id, severity)
obs.show_inputs(demo)
obs.show_rankings(ctx, demo)
```

Original: Tirasemtiv has no effect on fast-twitch muscle.
After typos: Tirasemtiv has no effect on fast-twitch musckle.

Maximum character edits: 1, actual edits: 1
muscle → musckle (insert)

Ranked papers before and after one character edit: Tirasemtiv has no effect on fast-twitch muscle. — dense top five, one query

Rank	Original query	Original cosine	After typos	Typo cosine
1	12631697 (judged relevant): Activation of fast skeletal muscle troponin as a potential...	0.380	23865182 (unjudged): Efficacy and tolerability of amitriptylinoxide in the...	0.308
2	43566999 (unjudged): Exercise blood flow patterns within and among rat muscles...	0.361	22995579 (unjudged): Amitriptyline is effective in chronic but not in episodic...	0.300
3	30101891 (unjudged): Intravenous valproate is well tolerated in unstable...	0.351	13624704 (unjudged): Ondansetron reduces nausea and vomiting after paediatric...	0.291
4	34328964 (unjudged): Intimal smooth muscle cell proliferation after balloon...	0.351	2475059 (unjudged): Once-a-day Concerta methylphenidate versus three-times-...	0.284
5	37037012 (unjudged): Regional vascular resistance during exercise: role of...	0.347	23783727 (unjudged): Ticagrelor vs. clopidogrel in patients with acute coronary...	0.276

Known relevant papers in these five results: 1 before edits → 0 after edits.

Dense nDCG@10: original 1.000 → after typos 0.000 (Δ -1.000); Recall@10: original 1.000 → after typos 0.000; finite RBO@10 (p=0.9): 0.025

5,183 documents, cached mode, original SciFact relevance judgments

Interpretation: After one character edit (muscle → musckle), only 3 of the 10 original papers stay in the dense top ten, and the known relevant paper is not among them, so nDCG@10 falls from 1.000 to 0.000.

Read the columns this way:

- `Rank` : position in dense search; 1 is first.
- `Original query` and `After typos` : papers returned for each input.
- The cosine columns: query–paper similarity; higher scores rank earlier.

Titles are shortened. Relevance labels use the same qrels as the first reference table.

The table shows five papers. nDCG, Recall, and RBO use ten. The next table compares all three methods before and after the typo.

```
# Compare original and typo rankings for the current settings.
obs.show_comparison(obs.observe(ctx, query_id, severity))
```

Query: Tirasemtiv has no effect on fast-twitch muscle. — one query, full collection, top 10 scored, RBO p=0.9, maximum/actual edits: 1/1

System	Original nDCG@10	After-typo nDCG@10	ΔnDCG@10	Finite RBO@10
BM25	1.000	0.000	-1.000	0.208
Dense	1.000	0.000	-1.000	0.025
Dense → reranker	1.000	0.000	-1.000	0.068

Original → typo nDCG@10 for each system: BM25 1.000 → 0.000; Dense 1.000 → 0.000; Dense → reranker 1.000 → 0.000.

Dense candidate Recall@20: original 1.000 → after typos 0.000; reranked Recall@10: original 1.000 → after typos 0.000 (both higher is better).

First changed dense rank: rank 1: 12631697 → 23865182 (original → after typos). The paper table shows five results; the scores use ten.

Interpretation: The typo lowers nDCG@10 for all three systems, and because no known relevant paper is left among the 20 dense candidates, the reranker has nothing to move up.

Find where the evidence was lost

Check the two recall values below the table:

- Candidate Recall@20=0: search found no known relevant paper. The reranker received no evidence.
- Reranked Recall@10 is lower than candidate Recall@20: some relevant candidates ended up below rank 10.

These checks separate missing candidates from papers ranked too low. BM25 shows how word matching handles the same typo.

Lesson Summary

First check whether search found the relevant paper. Then check where reranking placed it. Use qrels to measure relevance and RBO to compare rankings. M3 measures losses across all 300 test queries and tests more edits.

Next: M2: Diagnose query and corpus shifts.

Further reading

- Thakur et al. (NeurIPS 2021 Datasets and Benchmarks), BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models.
- Wadden et al. (EMNLP 2020), Fact or Fiction: Verifying Scientific Claims.

Appendix: How BM25 scores papers

BM25 stands for Best Matching 25. It adds up word-match scores and ranks papers from highest to lowest.

Each distinct query word contributes according to three rules:

- Rarity: a word found in fewer papers gets more weight (inverse document frequency, or IDF).
- Repetition: more occurrences in a paper raise its score, but each extra occurrence adds less.
- Length: at the same occurrence count, a longer paper gets less credit than a shorter one.

The implementation uses:

$$s(q,d) = \sum_{t \in Q} \mathrm{IDF}(t) \frac{f(t,d)(k_1+1)}{f(t,d) + k_1(1 - b + b|d|/\mathrm{avgdl})},$$

$$\mathrm{IDF}(t) = \ln\left(1 + \frac{N - df(t) + 0.5}{df(t) + 0.5}\right).$$

Here Q contains the distinct query words, N is the number of papers, $df(t)$ counts papers containing word t, and $f(t,d)$ counts its occurrences in paper d. $|d|$ is the paper's word count; avgdl is the average across all papers.

We use $k_1 = 1.2$ for repetition saturation and $b = 0.75$ for length correction. Text is the title plus abstract. We ignore case and split words at punctuation, so `fast-twitch` becomes `fast` and `twitch`; common words are kept and stemming is off.

For example, `muscle → musckle` removes the contribution from `muscle`: the misspelling does not occur in these papers.

Appendix: Calculate nDCG@10

1. Add credit for relevant papers

Let d_i be the paper at rank i. A known relevant SciFact paper gets $r_q(d_i) = 1$. An unjudged paper gets 0 for this calculation; this does not prove it is irrelevant.

The model score sets the order. The qrel sets the credit, called gain. The code uses linear gain in `tutorial_utils.metrics.ndcg`:

$$\mathrm{DCG@10}(q) = \sum_{i=1}^{10} \frac{r_q(d_i)}{\log_2(i+1)}.$$

A relevant paper adds 1 at rank 1, about 0.631 at rank 2, and about 0.289 at rank 10. Earlier is better. Papers below rank 10 add nothing.

2. Divide by the best possible score

IDCG@10 is the score of the best possible ranking. Put the highest qrel grades first and score the first ten. Include relevant papers that search missed. With m relevant papers, each with grade 1:

$$\mathrm{IDCG@10}(q) = \sum_{i=1}^{\min(m,10)} \frac{1}{\log_2(i+1)}, \qquad \mathrm{nDCG@10}(q) = \frac{\mathrm{DCG@10}(q)}{\mathrm{IDCG@10}(q)}.$$

IDCG stays fixed for the same query and qrels. A ranking with the best possible score has nDCG@10=1.

If a query has no positive qrel, IDCG is zero and nDCG is undefined. The code returns NaN, leaves it out of the mean, and reports the exclusion. Every supplied test query has a positive qrel, so none are excluded.

3. Example: Relevant papers at ranks 2 and 5

A query has two known relevant papers, A and B. The other results have no labels and get zero gain. Their relevance is unknown. Numbers in the table are rounded; the calculation uses full precision.

Rank i	Paper	Gain $r_q(d_i)$	Discount $1/\log_2(i+1)$	Contribution
1	U1	0	1.0000	0
2	A	1	0.6309	0.6309
3	U2	0	0.5000	0
4	U3	0	0.4307	0
5	B	1	0.3869	0.3869
6	U4	0	0.3562	0
7	U5	0	0.3333	0
8	U6	0	0.3155	0
9	U7	0	0.3010	0
10	U8	0	0.2891	0

Interpretation: Only the relevant papers A and B add to the score, and B adds less because it ranks later.

$$\mathrm{DCG@10} = \frac{1}{\log_2(3)} + \frac{1}{\log_2(6)} \approx 1.0178.$$

The best ranking puts A and B first and second:

$$\mathrm{IDCG@10} = 1 + \frac{1}{\log_2(3)} \approx 1.6309, \qquad \mathrm{nDCG@10} = \frac{1.0178\ldots}{1.6309\ldots} \approx 0.6241.$$

Recall@10=1 because both papers are in the top ten. nDCG@10 is below 1 because they could rank earlier.

If B moves to rank 15, it adds no credit. IDCG stays fixed, nDCG@10 falls to about 0.3869, and Recall@10 falls to 0.5.

4. Read the score

For a query with positive qrels, nDCG@10 ranges from 0 to 1. Higher is better.

nDCG@10	Meaning under the available judgments
1	The first ten results have the best possible score.
Between 0 and 1	Some known relevant papers are missing or rank later than in the best ranking.
0	No known relevant paper appears in the first ten.

Interpretation: nDCG rewards finding known relevant papers early, and a score of zero applies only to the first ten results; relevant papers may still appear later.

With exactly one known relevant paper:

Its rank	nDCG@10 ↑	Recall@10 ↑	Included at K=5?
1	1.0000	1	Yes
2	0.6309	1	Yes
5	0.3869	1	Yes
10	0.2891	1	No
11 or later, or absent	0	0	No

Interpretation: nDCG@10 falls with every step down while Recall@10 stays 1 inside the top ten, and a paper at rank 10 still counts toward both scores but stays out of a prompt with five references.

nDCG=0.6241 means the ranking earns 62.41% of the best possible score. Recall measures the fraction of relevant papers found.

If more than ten papers are relevant, nDCG@10 can be 1 while Recall@10 is below 1. M3 calculates nDCG for each query, then averages with equal weight for every query.